

ANIRUDH ARORA

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RESEARCH INTERESTS

Tropical precipitation and Indian Summer Monsoon dynamics in km-scale and SR-ESMs; representation of cloud microphysics and its impact on simulated precipitation; evaluation of CMIP-class models, large ensembles and the quantification of internal variability; climate model uncertainty and its translation into actionable, policy-relevant science.

EDUCATION

Indian Institute of Science Education and Research, Bhopal	2021 - 2026
<i>Integrated BS-MS (Dual Degree) in Environmental Sciences</i>	<i>Bhopal, MP</i>
Overall CPI: 9.22/10.00 ; Master's CPI: 10.00/10.00	

FORMAL RESEARCH EXPERIENCE

Max Planck Institute for Meteorology (MPI-M)	Supervisor - Dr. Clara Bayley
<i>July 2026 – November 2026</i>	<i>Hamburg, Germany</i>

- Research Assistant, Climate Surface Interaction Group, Climate Physics Department.
- Developing a convergence framework for the superdroplet model CLEO, quantifying how superdroplet resolution, stochastic sampling and initialization affect droplet-size distributions and bulk cloud-microphysics.

CLIMATics Group, KAUST	Supervisor - Prof. Ibrahim Hoteit
<i>July 2025 - June 2026</i>	<i>Thuwal, Saudi Arabia</i>

- Master's thesis: *Toward Reliable Tropical Precipitation Simulations by Assessing the Added Value of Large Ensembles and Kilometer-Scale Simulations.*
- Evaluated the IFS-FESOM nextGEMS production historical simulation at kilometre scale against IMD, GPCP, ERA5 and CMIP6 MMM benchmarks.
- Quantified ensemble convergence (N_{eff}) for CESM2-LE, MPI-GE and CESM2-PPE precipitation using four different criteria to assess the sample size needed to reliably constrain monsoon properties.
- Conducted the first dedicated long-term satellite-based analysis of atmospheric methane over the Arabian Peninsula using 93 months (May 2018 – December 2025) of bias-corrected blended TROPOMI+GOSAT XCH₄ retrievals. Manuscript in preparation for submission to *Atmospheric Chemistry and Physics* (ACP)

DST CoE for Climate Information, IIT Delhi	Supervisor - Prof. Saroj K. Mishra
<i>May 2023 - June 2026</i>	<i>New Delhi, India</i>

- Co-supervised by Prof. Mishra for my master's thesis.
- Evaluated CMIP5/CMIP6 multi-model ensemble and IFS-FESOM km-scale simulations of historical precipitation and Indian Summer Monsoon over the South Asian domain.
- Co-authored the SARCI framework for climate services in South Asia paper, published in *BAMS*.
- Investigated circulation-driven precipitation and temperature responses under solar geoengineering (GeoMIP G1 and G6sulfur), characterising regional monsoon and tropical rainfall changes.
- Co-developed a sequential decision-making (MDP/RL) framework for efficient climate action.
- Contributed to a multi-scale source attribution of natural and anthropogenic air pollution over the Indo-Gangetic Plain using reanalysis and satellite datasets.

Ethnography Studies - IISER Bhopal	Supervisor - Dr. Renny Thomas
<i>August 2023 - April 2024</i>	

- Conducted a field ethnography study to understand the socio-cultural importance of Garuda Purana, the book about death rituals and life after death.

- Conducted a comprehensive ethnographic study of two distinct laboratory settings over a four-month period, aiming to understand the socio-cultural dynamics, work practices, and interactions within these scientific environments.

PEER-REVIEWED PUBLICATIONS

- Bhuyan, D. P., Upadhyaya, P., Pathak, R., Namdev, P., Salunke, P., Anand, A., Suresh, A. D., **Arora, A.**, Baraik, S. K., Jain, S., Parihar, R. S., Dwivedi, A., Sahany, S., Sharan, M., Dash, S. K., Fasullo, J. T., Behera, S. K., Tribbia, J. and Mishra, S. K. (2026), *Advancing climate services in South Asia: The SARCI Framework for actionable information, capability, and capacity*. Bulletin of the American Meteorological Society, BAMS-D (DOI 10.1175/BAMS-D-25-0172.1).
- Mishra, S. K., Upadhyaya P., **Arora A.**, Vinod A.J., Jayan N., Dwivedi A. (2025), *To Solve North India's Air Crisis, Look Beyond Stubble and Fireworks*. The India Forum, **Oct 22, 2025**.

MANUSCRIPTS UNDER REVIEW

- **Arora, A.**, Dasari, H.P., Jha, S.K., Mishra, S.K., Hoteit, I. (2026), *Km-scale resolution improves monsoon precipitation characteristics but leaves large-scale dynamical biases: evidence from IFS-FESOM nextGEMS*. Under peer review at *Climate Dynamics*.
- **Arora, A.** (2026), *Public enlightenment and climate action: rethinking Kantian principles for a sustainable future*. Under peer review at *npj Climate Action*.

MANUSCRIPTS IN PREPARATION

- **Arora, A.**, Gandhi, H., Hoteit, I., *Methane variability, persistent hotspots, and shamal-driven seasonality over the Arabian Peninsula from blended TROPOMI+GOSAT XCH₄ retrievals (2018–2025)*. In preparation for *Atmospheric Chemistry and Physics*.
- **Arora, A.** & Upadhyaya, P., Mishra, S.K., Bhuyan, D., Behera, S., Fasullo, J., Tribbia, J., Jain, S., Salunke, P., Sahany, S., Dash, S.K., *Climate models doubled in a decade but credibility remains poor over the Global South*.
- Mishra, S.K., Ray, P., Srivastava, A., Upadhyaya, P., Ortiz, P., **Arora, A.**, Jain, S., Salunke, P., Schlosser, C., *A principled framework for sequential decision making to promote efficient climate actions*.

CONFERENCE CONTRIBUTIONS

- **Arora, A.**, Dasari, H.P., Jha, S.K., Mishra, S.K., Hoteit, I. (2026), *Evaluating Indian Summer Monsoon Precipitation in the IFS-FESOM nextGEMS Production Historical Simulation at Kilometre Scale*. km-Scale Global Modelling Summit 2026, Hamburg, Germany (Poster).
- **Arora, A.**, Upadhyaya, P., Mishra, S.K. (2024), *Circulation Changes under Geoengineering and their Implications in Regional Precipitation and Temperature*. 14th Annual GeoMIP Meeting 2024, Cornell University, USA. (Poster)

TECHNICAL SKILLS

Languages	Python, MATLAB, NCAR Command Language (NCL), Mathematica
Climate tools	Climate Data Operators (CDO), netCDF Operators (NCO), NetCDF
HPC & workflows	Slurm, Levante (DKRZ), Shaheen III (KAUST), Bash, Git
Typesetting & Web	L ^A T _E X, HTML5, CSS

HONORS

Rhodes Scholarship 2026 Finalist
University of Oxford – Rhodes Trust

November 2025

- Among the top 13 candidates out of ~1,000 applicants for the Rhodes Scholarship India.

Visiting Student Research Program*King Abdullah University of Science and Technology (KAUST)*

March 2025

Kingdom of Saudi Arabia

- Received full scholarship to conduct master's thesis research at KAUST under Prof. Ibrahim Hoteit.

Full Travel Support – 14th Annual GeoMIP Meeting*Cornell University*

July 2024

Ithaca, NY, USA

- Full travel and accommodation support to present at the 14th Annual GeoMIP Meeting.

CNR Rao Education Foundation Prize*Indian Institute of Science Education and Research*

August 2022

Bhopal, MP

- Awarded for the highest CPI (10.00/10.00) in the first semester of the BS-MS dual-degree programme.

Bronze Medal – The University Physics Competition*Carroll College, Helena, MT*

January 2023

- Team award in the worldwide University Physics Competition 2022 for a mathematical model of **perfect penalty kick trajectories**.

Earlier honours

- GIZ Sustainable Development Competition (felicitated at the German Embassy, New Delhi, 2019); Joseph Memorial Scholarship for academic excellence and leadership, St. Paul H.S. School, Indore (2019).

POSITIONS OF RESPONSIBILITY

Prayaas - Making Lives Colorful*Coordinator*

Student Organization

2022 - 2026

- Volunteer group working on the upliftment of underprivileged communities through education and social initiatives.

MU20 School of Opportunity*Executive Board Member*

Educational Venture

2022 - 2023

- Invited to serve as the Executive Board Member for a Model United Nation conference held at Indore World Summit 2022, one of central India's largest Education Conferences

IISER Bhopal Debating Society*President(2022) and Active member*

Student Organization

*2022 - 2024***Chrysalis IISER Bhopal***Podcast Creator*

Science Communication Body

2022

- Co-hosted a podcast on the topic, "Climate Change Research Governing Public Policy Making" and talked about the challenges and opportunities to integrate climate change research with public policymaking.